

Joint probability distributions

If x & y are two discrete random Variables we define the joint probability fcn of x & y by

$$P(X=x, Y=y) = f(x, y) \quad \text{Bivariate probability fcn}$$

Where $f(x, y)$ satisfy the cond^s $f(x, y) \geq 0$ & $\sum_x \sum_y f(x, y) = 1$ then f is referred as joint probability density fcn of x & y in the respective order

The set of values of this fcn $f(x_i, y_j) = J_{ij}$ for $i=1, 2, \dots$ & $j=1, 2, \dots$ is called joint probability distribution of x & y .

These values are presented in the form of a table called joint probability Table

$x \backslash y$	y_1	y_2	...	y_n	Sum
x_1	J_{11}	J_{12}	...	J_{1n}	$f(x_1)$
x_2	J_{21}	J_{22}	...	J_{2n}	$f(x_2)$
x_3	J_{31}	J_{32}	...	J_{3n}	$f(x_3)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_m	J_{m1}	J_{m2}	...	J_{mn}	$f(x_m)$
Sum	$g(y_1)$	$g(y_2)$	...	$g(y_n)$	1

$f(x_i)$ is the sum of entries of the 1st row, $f(x_2)$ is the sum of entries of the 2nd row & so on, $\dots$
 $g(y_1), g(y_2), \dots$ be the sum of entries of the 1st, 2nd, $\dots$ column respectively, then $\{f(x_1), f(x_2), \dots, f(x_m)\}$ & $\{g(y_1), g(y_2), \dots, g(y_n)\}$ are called the Marginal Probability distributions.

$$\sum_{i=1}^m \sum_{j=1}^n f(x_i, y_j) = \sum_{i=1}^m \sum_{j=1}^n J_{ij} = 1$$

Expectation, Variance, Co-variance & Correlation

If X & Y are RVs having joint probability fcn $f(x, y)$, then the Expectation of X & Y is denoted by $E(X, Y)$ @ μ_{xy} is defined by the relation

$$E(XY) = \sum_{i,j} x_i y_j f_{ij}$$
$$\mu_x = E(X) = \sum_i \sum_j x_i f(x_i, y_j) = \sum_i x_i f(x_i)$$
$$\mu_y = E(Y) = \sum_i \sum_j y_j f(x_i, y_j) = \sum_j y_j g(y_j)$$

If $z = \phi(x, y)$ & $f(x, y)$ is the joint distribution of X & Y , the Expectation of z in the joint distribution of X, Y is defined by the relation

$$E(z) = \sum_i \sum_j \phi(x_i, y_j) f_{ij}$$

If X & Y are random variables having mean μ_x & μ_y respectively, then the Covariance of X & Y denoted by $\text{Cov}(X, Y)$ is defined by the relation

$$\text{Cov}(X, Y) = \sum_i \sum_j (x_i - \mu_x)(y_j - \mu_y) f_{ij}$$

$$\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$$

$$\textcircled{2} \text{Cov}(X, Y) = E(XY) - \mu_x \mu_y = E(XY) - E(X)E(Y)$$

Also Correlation = $\rho(x, y) = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$

Note:- If X & Y are independent random variable, then

$$E(XY) = E(X) \cdot E(Y)$$

$$\text{Cov}(X, Y) = 0 \text{ \& \; hence } E(XY) = 0$$

$$\sigma_x^2 = E(X^2) - [E(X)]^2$$

$$\sigma_y^2 = E(Y^2) - [E(Y)]^2$$

The discrete random variables X & Y are said to be independent if

$$P(X=x, Y=y) = P(X)P(Y)$$

i.e

$$\text{i.e } f(x) \cdot g(y) = f_{ij}$$

The joint probability distribution of two variables X & Y is as follows

$X \backslash Y$	-4	2	7
1	$\frac{1}{8} J_{11}$	$\frac{1}{4} J_{12}$	$\frac{1}{8} J_{13}$
5	$\frac{1}{4} J_{21}$	$\frac{1}{8} J_{22}$	$\frac{1}{8} J_{23}$

Compute

- (i) $E(X)$ & $E(Y)$ (ii) $E(XY)$
 (iii) σ_x & σ_y (iv) $\text{cov}(X, Y)$

soln:- The distribution is obtained by adding all the respective row entries & the column entries

Distribution of X

$$x_i: \quad 1 \quad 5$$

$$f(x_i): \quad \frac{1}{2} \quad \frac{1}{2}$$

Distribution of Y

$$y_i: \quad -4 \quad 2 \quad 7$$

$$f(y_i): \quad \frac{3}{8} \quad \frac{3}{8} \quad \frac{1}{4}$$

$$(i) E(X) = \sum x f(x) = (1) \frac{1}{2} + 5 \left(\frac{1}{2} \right) = 3 = \mu_x$$

$$E(Y) = \sum y f(y) = (-4) \left(\frac{3}{8} \right) + 2 \left(\frac{3}{8} \right) + 7 \left(\frac{1}{4} \right) = 1 = \mu_y$$

$$(ii) E(XY) = \sum_i \sum_j x_i y_j J_{ij}$$

$$= (-4) \left(\frac{1}{8} \right) + (1)(2) \left(\frac{1}{4} \right) + (1)(7) \left(\frac{1}{8} \right) + 5(-4) \left(\frac{1}{4} \right) + 5(2) \left(\frac{1}{8} \right) + 5(7) \left(\frac{1}{8} \right)$$

$$E(X^2) = \frac{9}{2}$$

$$E(X^2) - [E(X)]^2$$

$$(iii) \sigma_x^2 = E(X^2) - \mu_x^2 \quad \& \quad \sigma_y^2 = E(Y^2) - \mu_y^2$$

$$\text{Now } E(X^2) = \sum x_i^2 f(x_i) = 1 \left(\frac{1}{2} \right) + 25 \left(\frac{1}{2} \right) = 13$$

$$\therefore \sigma_x^2 = 13 - 3^2 = 13 - 9 = 4 \Rightarrow \boxed{\sigma_x = 2}$$

$$\text{So } E(Y^2) = \sum y_i^2 f(y_i) = (-4)^2 \left(\frac{3}{8} \right) + 4 \left(\frac{3}{8} \right) + 49 \left(\frac{1}{4} \right) = \frac{79}{4}$$

$$\sigma_y^2 = \frac{79}{4} - 1^2 = \frac{75}{4} \Rightarrow \boxed{\sigma_y = 4.33}$$

$$(iv) \text{Cov}(X, Y) = E(XY) - \mu_x \mu_y$$

$$= \left(\frac{3}{2}\right) - (3)(1) = -3/2$$

$$\boxed{\text{Cov}(X, Y) = -3/2}$$

(2) A joint probability fn is given by the tbl table
Find $\text{cov}(X, Y)$ & $\rho(X, Y)$

$X \backslash Y$	-3	2	4
1	0.1	0.2	0.2
3	0.3	0.1	0.1

Soln:- Distribution of X

$$x_i: \quad 1 \quad 3$$

$$f(x_i): \quad 0.5 \quad 0.5$$

Distribution of Y

$$y_i: \quad -3 \quad 2 \quad 4$$

$$f(y_i): \quad 0.4 \quad 0.3 \quad 0.3$$

We have $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$

$$E(X) = \sum x_i f(x_i) = 1(0.5) + 3(0.5) = 2$$

$$E(Y) = \sum y_i f(y_i) = -3(0.4) + 2(0.3) + 4(0.3) = 0.6$$

$$E(XY) = \sum_{i,j} x_i y_i f_{ij} = (1)(-3)(0.1) + (3)(2)(0.2) + (1)(4)(0.2) + (3)(-3)(0.3) + 3(2)(0.1) + 3(4)(0.1)$$

$$E(XY) = 0$$

$$\therefore \text{Cov}(X, Y) = 0 - (2)(0.6) = -1.2$$

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{-1.2}{(2)(0.6)} = -1 \quad \boxed{-1}$$

1. The joint distribution of two random Variables X & Y is given by the following table

x_j 1	2	3	4
1	0.06	0.15	0.09
2	0.14	0.35	0.21

Determine the marginal distribution of X & Y . Also verify that X & Y are independent

Soln:- The marginal distributions of X & Y are

$$x_i: 1 \quad 2$$

$$f(x_i): 0.3 \quad 0.7$$

$$y_i: 2 \quad 3 \quad 4$$

$$g(y_i): 0.20 \quad 0.50 \quad 0.30$$

To Verify X & Y are independent

The Variables X & Y are independent if

$$f(x_i) g(y_i) = J_{ij}$$

$$f(x_1) g(y_1) = (0.3)(0.2) = 0.06 = J_{11}$$

$$f(x_2) g(y_2) = (0.7)(0.5) = 0.35 = J_{22}$$

Hence X & Y are independent

If X & Y are two discrete R.V. having joint prob. fcn $f(x, y) = k(2x + y)$, where x & y are integers satisfying $0 \leq x \leq 2$, $0 \leq y \leq 3$. Find the value of the constants k & $E(X)$, $E(Y)$, $E(X^2)$, $E(Y^2)$.

Solⁿ:- $X = \{0, 1, 2\}$, $Y = \{0, 1, 2, 3\}$

$$f(x, y) = k(2x + y)$$

$x \backslash y$	0	1	2	3	Sum
0	0	k	$2k$	$3k$	$6k$
1	$2k$	$3k$	$4k$	$5k$	$14k$
2	$4k$	$5k$	$6k$	$7k$	$22k$
Sum	$6k$	$9k$	$12k$	$15k$	$42k$

$$6k + 14k + 22k = 1$$

$$42k = 1$$

$$k = \frac{1}{42}$$

$$\therefore k = \frac{1}{42} \quad \because \sum_j f_{ij} = 1$$

$\therefore$ The Marginal Prob. distribution is given as

$$x : 0 \quad 1 \quad 2$$

$$f(x) : \frac{6}{42} \quad \frac{14}{42} \quad \frac{22}{42}$$

$$y : 0 \quad 1 \quad 2 \quad 3$$

$$g(y) : \frac{6}{42} \quad \frac{9}{42} \quad \frac{12}{42} \quad \frac{15}{42}$$

$$x : 0 \quad 1 \quad 2$$

$$f(x) : \frac{1}{7}$$

Coin is tossed three times. let X denote ϕ or 0 according as a tail or a head occurs on first toss. let Y denote the no of heads which occurs. Determine (i) the distributions of X & Y (ii) The joint distribution of X & Y find the expected values of $X+Y$ & XY

Soln, $S = \{HHH, HHT, HTH, HTT, TTH, THT, TTT\}$

$X: \{0, 0, 0, 0, 1, 1, 1\}$

$Y: \{3, 2, 2, 1, 2, 1, 0\}$

(i) $\therefore X = \{0, 1\}$, $Y = \{0, 1, 2, 3\}$

$X \backslash Y$	0	1	2	3	$f(x)$
0	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{1}{2}$
1	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0	$\frac{1}{2}$
$g(y)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	$\sum = 1$

(ii) Marginal distribution of X

$X: 0, 1$

$f(x): \frac{1}{2}, \frac{1}{2}$

Marginal distribution of Y

$Y: 0, 1, 2, 3$

$g(y): \frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$

(iii) $Z = X+Y = x_i + y_j = \{0, 1, 2, 3, 4\}$

Corresponding probabilities are

$\{0, \frac{1}{8}, \frac{2}{8} + \frac{2}{8}, \frac{1}{8} + \frac{1}{8}, 0\}$

$P(Z) = \{0, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 0\}$

$\therefore$ $X+Y: 0, 1, 2, 3$

$P(X+Y): 0, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}$

$\therefore E[X+Y] = \sum (x+y) \cdot P(X+Y) = 0 + \frac{1}{4} + 1 + \frac{3}{4} = 1$

If X & Y are independent R.V. X takes values 2, 5, 7 with prob. $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$ respectively. Y takes values 3, 4, 5 with prob. $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$

(i) Find the Prob. distribution of X & Y (ii) S.T the Covariance of X & Y is equal to zero (iii) Find the prob. distrib. of $Z = X + Y$

Soln

$X : \begin{matrix} x_1 & x_2 & x_3 \\ 2 & 5 & 7 \end{matrix}$ $Y : \begin{matrix} y_1 & y_2 & y_3 \\ 3 & 4 & 5 \end{matrix}$

$P(X) : \frac{1}{2} \quad \frac{1}{4} \quad \frac{1}{4}$ $g(Y) : \frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{3}$

(i)

$X \backslash Y$	3	4	5	$f(x)$
2	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{2}$
5	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$
7	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$
$g(y)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\sum f(x) = 1$

S.T X & Y are independent
 $f_{ij} = f(x_i) g(y_j)$

(ii) $Cov(X, Y) = E(XY) - E(X)E(Y)$

$E(X) = \sum x_i f(x_i) = 4$

$E(Y) = \sum y_j g(y_j) = 4$

$\therefore Cov(X, Y) = 0$

$E(XY) = \sum x_i y_j f_{ij} = 16$

$\{5, 6, 7, 8, 9, 10, 10, 11, 12\}$

(iii) $Z = X + Y$

$Z_i = x_i + y_j = \{Z_i\} = \{5, 6, 7, 8, 9, 10, 11, 12\}$

The corresponding probabilities are.

$P(Z_i) = \left\{ \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12} + \frac{1}{12}, \frac{1}{12}, \frac{1}{12} \right\}$

$\therefore$ The prob. distribution of $Z = X + Y$ be

Z	5	6	7	8	9	10	11	12
$P(Z)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{12}$

Two Cards are selected at a random from a box which contains 5 cards numbered 1, 1, 2, 2 & 3. Find the joint distribution of X & Y where X denoted the sum & Y the maximum of the two numbers drawn. Also determine $\text{Cov}(X, Y)$ & $\rho(X, Y)$

Soln:- The possible pair of nos are

$$S = \{ (1, 1) (1, 2) (1, 2) (1, 3) (1, 2) (1, 2) (1, 3) (2, 2) (2, 3) (2, 3) \}$$

$$X: \text{Sum of two nos}$$

$$Y: \text{Max of the two numbers drawn}$$

$$X: 2, 3, 4, 5$$

$$Y: 1, 2, 3$$

Joint prob. distribution, table be

$X \backslash Y$	1	2	3	$g(x)$
2	$1/10$	0	0	0.1
3	0	$4/10$	0	0.4
4	0	$1/10$	$2/10$	0.3
5	0	0	$2/10$	0.2
$g(y)$	0.1	0.5	0.4	$\sum_{i,j} = 1$

Distribution on X

$$X: 2 \quad 3 \quad 4 \quad 5$$

$$f(x): 0.1 \quad 0.4 \quad 0.3 \quad 0.2$$

Distribution on Y

$$Y: 1 \quad 2 \quad 3$$

$$g(y): 0.1 \quad 0.5 \quad 0.4$$

$$\text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$$

$$E(X) = \sum_{i=1}^4 x_i f(x_i) = 2(0.1) + 3(0.4) + 4(0.3) + 5(0.2) = \underline{3.6}$$

$$E(Y) = \sum_{j=1}^3 y_j g(y_j) = 1(0.1) + 2(0.5) + 3(0.4) = \underline{2.3}$$

$$E(XY) = \sum \sum x_i y_j T_{ij} = 2(1)(0.1) + 2(2)(0) + \dots + 3(2)(0.4) + \dots$$

$$+ 4(2)(0.1) + 4(3)(0.2) + \dots + 5(3)(0.2) = \underline{8.8}$$

$$\text{Cov}(x, y) = 8.8 - (2.6)(2.3) = \boxed{0.52 = \text{Cov}(x, y)}$$

$$\rho(x, y) = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$\sigma_x^2 = E(x^2) - (E(x))^2$$

$$\sigma_x^2 = \sum x^2 f(x) - (2.6)^2 = 0.84$$

$$\sigma_y^2 = E(y^2) - (E(y))^2$$

$$\sigma_y^2 = 5.7 - (2.3)^2 = 0.41$$

$$\therefore \rho(x, y) = \frac{0.52}{(0.9165)(0.6402)} = 0.886 \Rightarrow \boxed{\rho(x, y) = 0.89}$$

Joint Probability Distribution of Continuous R.V

Let X & Y be two Continuous Random Variables
If $f(x, y)$ is a ~~pdf~~ real valued ~~fn~~ satisfying the conditions

$$f(x, y) \geq 0 \quad \& \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

then $f(x, y)$ is called the joint pdf. of the random Variable X & Y

$$P(a \leq x \leq b, c \leq y \leq d) = \int_a^b \int_c^d f(x, y) dy dx$$

Expectation, Variance & Covariance:

$$E(X) = \mu_x = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy$$

$$E(Y) = \mu_y = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy$$

$$\sigma_x^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_x)^2 f(x, y) dx dy = E[(X - \mu_x)^2]$$

$$\sigma_y^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y - \mu_y)^2 f(x, y) dx dy = E[(Y - \mu_y)^2]$$

$$\text{Cov}(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (X - \mu_x)(Y - \mu_y) f(x, y) dx dy$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

Note: ① If X & Y are independent, then

$$E(XY) = E(X) \cdot E(Y)$$

$$\text{then } \text{Cov}(X, Y) = 0$$

② The marginal distribution of X & Y are $g(x)$ & $h(y)$, which are given by

$$g(x) = \int_{y=-\infty}^{\infty} f(x, y) dy \quad \& \quad h(y) = \int_{x=-\infty}^{\infty} f(x, y) dx$$

③ The conditional probability of x given y is given by

$$g(x/y) = \frac{f(x, y)}{h(y)} \quad h(y) > 0 \quad \&$$

$$h(y/x) = \frac{f(x, y)}{g(x)} \quad \dots \quad g(x) > 0$$

① Find k so that $P(x,y) = \begin{cases} k(x+1)e^{-y} & 0 < x < 1, y > 0 \\ 0 & \text{otherwise} \end{cases}$ is a joint pdf

Soln:-

$$(i) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$$

$$\int_0^1 \int_0^{\infty} k(x+1)e^{-y} dy dx = 1$$

$$\Rightarrow \int_0^1 \left[\frac{k(x+1)e^{-y}}{-1} \right]_0^{\infty} dx = 1$$

$$\Rightarrow \int_0^1 -k(x+1)[0-1] dx = 1$$

$$\Rightarrow \int_0^1 k(x+1) dx = 1$$

$$\Rightarrow \left[\frac{kx^2}{2} + x \right]_0^1 = 1$$

$$\Rightarrow \frac{k}{2} + 1 - 0 = 1$$

$$\Rightarrow \frac{k}{2} = 0$$

$$\Rightarrow \boxed{k=0}$$

② If the joint Prob. f_{xy} of the random variable X & Y is given by $f(x,y) = \begin{cases} \frac{1}{8}(6-x-y) & 0 < x < 2, 2 < y < 4 \\ 0 & \text{otherwise} \end{cases}$

Find (a) $P(x < 1, y < 3)$ (b) $P(x+y < 3)$

Soln:-

$$P(x < 1, y < 3) = \int_{x=0}^1 \int_{y=2}^3 \frac{1}{8}(6-x-y) dy dx$$

$$= \frac{1}{8} \int_{x=0}^1 \left[6y - xy - \frac{y^2}{2} \right]_2^3 dx$$

$$= \frac{1}{8} \int_{x=0}^1 \left(6 - x - \frac{5}{2} \right) dx$$

$$= \frac{1}{8} \left[6x - \frac{x^2}{2} - \frac{5x}{2} \right]_0^1 =$$

$$P(X < 1, Y < 3) = \frac{3}{8}$$

$$(6) P(X+Y < 3) = \int_{x=0}^1 \int_{y=2}^{3-x} \frac{1}{8} (6-x-y) dy dx$$

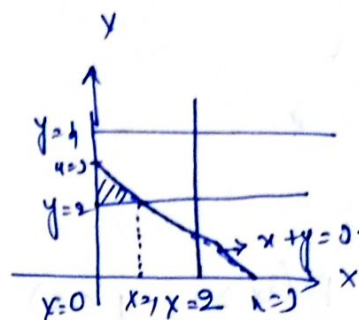
$$= \frac{1}{8} \left[\int_0^1 \left(6y - xy - \frac{y^2}{2} \right)_{y=2}^{3-x} dx \right]$$

$$= \frac{1}{8} \left[\int_0^1 \left(6[1-x] - x[1-x] - \frac{1}{2} [(3-x)^2 + 2] \right) dx \right]$$

$$= \frac{1}{16} \int_0^1 (x^2 - 8x + 7) dx$$

$$= \frac{1}{16} \left[\frac{x^3}{3} - 4x^2 + 7x \right]$$

$$P(X+Y < 3) = \frac{5}{24} = 0.2083$$



(3) Find the value of the const C , s.t
 $f(x,y) = \begin{cases} C(2x+y), & 0 \leq x \leq 2, 0 \leq y \leq 3, \\ 0 & \text{otherwise} \end{cases}$ is a joint

pdf of x & y . Hence Evaluate $P(x \geq 1, y \leq 2)$

Soln:- We must have

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$$

$$\Rightarrow \int_{x=0}^2 \int_{y=0}^3 C(2x+y) dy dx = 1$$

$$\Rightarrow C \int_0^2 \left(2xy + \frac{y^2}{2} \right)_{y=0}^3 dx = 1$$

$$\Rightarrow C \int_0^2 \left(6x + \frac{9}{2} \right) dx = 1$$

$$\Rightarrow C \left(3x^2 + \frac{9x}{2} \right)_0^2 = 1 \Rightarrow C(12+9) = 1$$

$$\Rightarrow \boxed{C = \frac{1}{21}}$$

$$P(x \geq 1, y \leq 2) = \int_{x=1}^2 \int_{y=0}^2 \frac{1}{21} (2x+y) dy dx$$

$$= \frac{1}{21} \int_{x=1}^2 \left[2x+y + \frac{y^2}{2} \right]_0^2 dx = \frac{1}{21} \int_0^2 (4x+2) dx$$

$$= \frac{1}{21} \left[2x^2 + 2x \right]_1^2$$

$$P(x \geq 1, y \leq 2) = \underline{\underline{\frac{8}{21}}}$$

(A) If X & Y are random Variables having joint pdf
 $f(x,y) = \begin{cases} 4xy, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$ Verify that

(i) $E(X+Y) = E(X) + E(Y)$

(ii) $E(XY) = E(X) \cdot E(Y)$

Soln:-

$$E(X) = \mu_x = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x,y) dx dy$$

$$= \int_{y=0}^1 \int_{x=0}^1 x \cdot 4xy dy dx = \int_{y=0}^1 \left[4x^2 \frac{y^2}{2} \right]_0^1 dx$$

$$E(X) = \int_{y=0}^1 2x^2 dx = \left(2 \frac{x^3}{3} \right)_0^1 \Rightarrow \frac{2}{3} //$$

$$\boxed{E(X) = \frac{2}{3}}$$

$$E(Y) = \mu_y = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x,y) dx dy = \int_0^1 \int_0^1 y \cdot 4xy dx dy$$

$$E(Y) = 4 \int_{x=0}^1 4x \left(\frac{y^3}{3} \right)_0^1 dx = \frac{4}{3} \int_0^1 x dx = \frac{4}{3} \left(\frac{x^2}{2} \right)_0^1$$

$$\boxed{E(Y) = \frac{2}{3}}$$

$$E(X+Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f(x,y) dx dy$$

$$E(X+Y) = \frac{4}{3}$$

$$E(XY) = E(X) \cdot E(Y) = \frac{4}{9}$$

4) $f(x, y) = \begin{cases} x^2 + \frac{xy}{3} & ; 0 < x < 1, 0 < y < 2 \\ 0 & ; \text{otherwise} \end{cases}$ find
 (i) $P(X > \frac{1}{2})$
 (ii) $P(Y/x)$

Soln:- $P(X > \frac{1}{2}) = \int_{x=\frac{1}{2}}^1 \int_{y=0}^2 \left(x^2 + \frac{xy}{3} \right) dy dx$

$$P(Y/x) = \frac{f(x, y)}{g(x)} = \frac{f(x, y)}{\int_{y=0}^2 \left(x^2 + \frac{xy}{3} \right) dy} = \frac{x^2 + \frac{xy}{3}}{\int_{y=0}^2 \left(x^2 + \frac{xy}{3} \right) dy}$$

$$P(Y/x) = \frac{x^2 + \frac{xy}{3}}{\left(x^2 y + \frac{xy^2}{6} \right)_0^2} = \frac{x^2 + \frac{xy}{3}}{2x^2 + \frac{2xy}{3}} = \frac{x^2 + \frac{xy}{3}}{2 \left(x^2 + \frac{xy}{3} \right)}$$

$$\textcircled{1} f(x, y) = \begin{cases} c(x^2 + y^2), & 0 \leq x \leq 4, 0 \leq y \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

find (i) c (ii) $P(1 < x < 2, 2 < y < 3)$

(iii) $P(x \geq 3, y \leq 2)$ (iv)

Soln: WKT $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

$$\therefore \int_{x=0}^4 \int_{y=0}^4 c(x^2 + y^2) dy dx = 1$$

$$\Rightarrow c \int_{x=0}^4 \left(x^2 y + \frac{y^3}{3} \right) \Big|_0^4 dx = 1$$

$$\Rightarrow c \int_{x=0}^4 \left[4x^2 + \frac{64}{3} \right] dx = 1$$

$$\Rightarrow c \left(\frac{4x^3}{3} + \frac{64}{3}x \right) \Big|_0^4 = 1$$

$$\Rightarrow \frac{c}{3} [64(4) + 64(4)] = 1$$

$$\Rightarrow \boxed{c = \frac{3}{512}}$$

$$(i) P(1 < x < 2, 2 < y < 3) = \int_{x=1}^2 \int_{y=2}^3 c(x^2 + y^2) dy dx =$$

$P =$

A bag contains 3 white, 2 red, 2 green bulbs. 2 bulbs are selected at random, if X & Y are RV denoting no of white & red bulbs respectively, Determine (i) joint distribution of X & Y (ii) Marginal distribution of X & Y (iii) $Cov(X, Y)$

Sol:- $X : \{0, 1, 2, 3\} \rightarrow$ white $\begin{matrix} 3(0) \\ 2(2) \\ 2(2) \end{matrix}$
 $Y : \{0, 1, 2\} \rightarrow$ red

$X \backslash Y$	0	1	2
0	$\frac{1}{21}$	$\frac{4}{21}$	$\frac{1}{21}$
1	$\frac{6}{21}$	$\frac{6}{21}$	0
2	$\frac{3}{21}$	0	0
3	0	0	0

$$P(0,0) = \frac{{}^2C_2}{{}^7C_2} = \frac{1}{21}$$

$$P(0,1) = \frac{{}^2C_1 \times {}^2C_1}{{}^7C_2} = \frac{4}{21}$$

$$P(0,2) = \frac{{}^2C_2}{{}^7C_2} = \frac{1}{21}$$

$$P(1,0) = \frac{{}^3C_1 \times {}^2C_1}{{}^7C_2} = \frac{6}{21}$$

$$P(1,1) = \frac{{}^3C_1 \times {}^2C_1}{{}^7C_2} = \frac{6}{21}$$

$$P(2,0) = \frac{{}^3C_2}{{}^7C_2} = \frac{3}{21}$$

Marginal distributions

$X : 0 \quad 1 \quad 2 \quad 3$
 $f(x) : \frac{6}{21} \quad \frac{12}{21} \quad \frac{3}{21} \quad 0$

Marginal distribution on Y

$X : 0 \quad 1 \quad 2$
 $g(y) : \frac{10}{21} \quad \frac{10}{21} \quad \frac{1}{21}$

The conditional probability: Distribution:

The conditional probability of Random Variable $Y=y$ given $X=x$ is defined as

$$h(Y/X=x) = \frac{h(x,y)}{f(x)}, \quad f(x) > 0$$

The conditional probability of R.V. X given $Y=y$ is defined as

$$h(X/Y=y) = \frac{h(x,y)}{g(y)}, \quad g(y) > 0$$

(i) Given the bivariate P.d, obtain

(i) Marginal distribution of X

(ii) The conditional dist. of Y given $X=2$

(iii) ———— X given $Y=-1$

$X \backslash Y$	-1	0	1
0	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$
1	$\frac{3}{15}$	$\frac{2}{15}$	$\frac{1}{15}$
2	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$

Soln. (i) Marginal distributions of X & Y are

$X: \quad 0 \quad 1 \quad 2$

$Y: \quad -1 \quad 0 \quad 1$

$f(x): \quad \frac{4}{15} \quad \frac{6}{15} \quad \frac{5}{15}$

$g(y): \quad \frac{6}{15} \quad \frac{5}{15} \quad \frac{4}{15}$

$$(ii) \quad h(Y/X=2) = \frac{P(2, Y)}{f(2)} = \frac{\cancel{15}^2 P(2, Y)}{\cancel{15}}$$

$Y:$	-1	0	1
$h(Y/2)$	$3\left(\frac{2}{15}\right)$	$3\left(\frac{1}{15}\right)$	$3\left(\frac{2}{15}\right)$

$$(iii) \quad h(X/Y=-1) = \frac{P(X, -1)}{g(-1)} = \frac{\cancel{15}^5 P(X, -1)}{\cancel{15}_2} = \frac{5 P(X, -1)}{2}$$

X	0	1	2
$h(X/-1)$	$\frac{5}{2} P(0, -1)$	$\frac{5}{2} P(1, -1)$	$\frac{5}{2} P(2, -1)$
	$\frac{5}{2} \left(\frac{1}{15}\right)$	$\frac{5}{2} \left(\frac{3}{15}\right)$	$\frac{5}{2} \left(\frac{2}{15}\right)$